

research

Phase 0 — Research

Feature: Session record per chapter + QA readiness · **Date:** 2026-09-08

Every decision was settled by a measurement taken on this tree, quoted with it. Where a decision rests on judgement rather than measurement, that is said.

No NEEDS CLARIFICATION entered this phase — four were closed in the spec’s Clarifications section. What follows are the design decisions those did not settle, plus three refuted hypotheses kept because each pointed the work in a wrong direction before being killed.

R1 — Where does the session record’s data come from?

Decision: From `pipeline/extract/meeting_notes.py`, which already produced all four sections into `curriculum/chapter-<slug>/knowledge/*.jsonl`. Build the plan, the cumulative view, carry-forward and serving **around** it.

Rationale. The producer existed and was wired into the pipeline. The initial brief said “0 matches inside chapters/”, which was accurate and misleading — the material lives under `curriculum/`. Forking the producer would have created two extraction mechanisms for one record (§11.4.251) and, worse, two mechanisms that could disagree about the same section.

What was genuinely absent: the plan, the cumulative view, carry-forward, the empty-versus-unwritten distinction, and serving.

Alternatives considered: authoring the sections by hand (rejected — a hand-maintained list drifts from what the pipeline produces); a second extractor tuned for this feature (rejected — the fork).

R2 — How is carry-forward computed so a point cannot vanish?

Decision: `plan(N)` = unresolved points of `plan(N-1)` with `slip+1` **plus** chapter `N-1`'s next-meeting points. Coverage is a recorded *input*, and **absence of a decision resolves to carried**.

Rationale, and it is the load-bearing design choice. Resolve has **no fourth branch**. A planned point is covered, carried, or dropped-with-a-reason; there is no path by which it can be neither. **vanished=0 therefore holds by construction, not by diligence** — which is the only way it can hold across a corpus nobody re-reads.

A drop recorded **without a reason is refused** and carries instead, so “dropped” cannot become a quiet disposal route.

Ordering uses `curriculum.ChapterID.OrdinalPath`, which already exists. **A second ordering is a second thing to disagree.**

Measured: 66 planned, 0 covered, 66 carried, 0 dropped, **0 vanished**; 436 items, 210 carried, max slip 2, one terminal outbox holding 42 points.

R3 — Why is covered: 0, and why was a verdict refused?

Decision: `coverage_state`: undecided, and **no coverage verdict was authored**.

Rationale, in the deriving agent's own words: judging coverage means matching one heuristic extraction against another, and a text-match verdict would be “*a guess wearing a decision's clothes*” — every downstream percentage would inherit it.

The plan basis is weaker than an agenda and the record says so: it is a **cue-phrase heuristic over ASR text** whose own docstring admits it misses paraphrases. It is *what was said about the next session*, never an *agreed agenda*.

Chapter 01's plan is UNDETERMINED — no predecessor, no recorded intent anywhere. No plan was invented and **no coverage percentage is computed for it**.

The operator has since asked for a machine PROPOSAL under review. FR-007a is what makes that compatible with the refusal rather than an override: a proposal carries per-point evidence and confidence, is marked PROPOSED, and **the served coverage_state must remain undecided** until a human confirms. A draft must not move a published figure.

R4 — Where do the four sections live on the wire?

Decision: Both — a `session_record` key on GET `/api/chapters/{id}`, and three dedicated routes. The fourth is blocked.

Rationale. The embedded key was built first, deliberately avoiding route-manifest churn, and it works: four sections, 126 carry-forward markers, 281 slip references, one outbox. But **four Angular components already existed** for standalone routes, and the manifest carried four NOT_BUILT DEBT rows. The data was reachable one way while the UI was built for another.

The operator chose to build the routes. The binding constraint is that both surfaces **consume pkg/sessionrecord** rather than re-deriving — and the gate must assert they agree, because *a route and a key that agree today are exactly what drift*.

meeting-notes is NOT built. The manifest records that its withholding rule **may not be guessed**, because one note is a content judgement left undecided for the operator. FR-005a: a route built on a guessed withholding rule publishes a judgement nobody made. That refusal is preserved, not routed around.

R5 — REFUTED: “the platform is close to QA-ready”

Refuted by an adversarial independent review, and recorded because the wrong belief nearly shipped.

Four headline claims were false against the running product:

claimed	served
30 areas with an assessment	14
224 graded questions	116
lessons carry real content	0 of 318
four session sections served	0

The evidence machinery was never bluffing about what it measures. It was silent about what nothing measured: no gate checked bank-to-area reachability across the population, no gate asserted a lesson body is non-empty, and the server-unity gate exited **0** while printing four NOT_BUILT routes as recorded debt — honest in its own output, but a “PASS” anyone could quote.

Every one of the four sat in a blind spot. That is why each fix ships the missing gate alongside it, and why the constitution gained *A Gate’s Population Is Part of Its Claim*.

R6 — REFUTED: “the catalog loader is broken”

The 13 unreachable banks looked like a loader defect. **It is not.**

learning.Load reads the catalog **once at process start**, deliberately and documentedly. The server then answered “*this area’s learning catalog carries no end-of-area test*” — a claim about the **directory** — from a snapshot taken hours earlier.

The discriminating condition was time, not the areas and not the banks: the 14 that worked had banks written at 07:13; the 13 that failed at 10:47; the process started at 08:01.

Fix: the snapshot may only assert what it licenses. Staleness is four-valued — Fresh / Stale / StalenessUnknown / NoSource. NoSource is separate on purpose: an in-memory test catalog has no directory to have drifted from, so its negatives still stand. A stale catalog returns **503** naming the changed files.

Alternatives considered: reload on every request (rejected — solves the symptom at a cost the design deliberately avoided); watch the directory (rejected — more machinery than the honesty requires).

R7 — REFUTED: “the lesson bodies were lost”

They were **read, measured, and discarded** on every run. `build_learning_catalog.py` parsed every section body, used it once for `estimatedMinutes`, and never referenced it again.

The reason is the interesting part: the shared library’s Lesson had nowhere to put it. Its only text field was `Summary`, a one-line blurb the interface renders as a muted paragraph. Writing a 4,563-word body with time links into a field named `Summary` would have been a semantic lie, **so the builder correctly dropped it rather than lying.**

Therefore the fix belonged upstream, in `curriculum-kit`, not in the consumer — and a consumer-local field would have failed the **entire directory load**, because `DecodeDocument` calls `DisallowUnknownFields()`.

Body is **optional** in the kit by deliberate choice: a materials-only lesson is legitimate, so “every lesson must have a body” is a *consumer policy* the library may not impose. A test records that choice so a future rule requiring one must change a test that explains its absence.

Measured: 339 of 339 at source, **318 of 318 served**, median 328 words, longest 4,563. Twelve lessons under 100 words were **left short** — converted, not padded.

R8 — How is the answer key kept off the wire?

Decision: A QuestionForTaking projection built as a map with **no answer field at all** — not the domain type with fields omitted.

Rationale. The kit's Question carries CorrectChoices/Explanation with omitempty, so **marshalling the type directly would ship the key** the moment a question had one. Omission depends on every future field remembering to omit itself; **absence does not**.

And the lock was bypassable one route over. The practice deck and the graded bank are the same authored rows emitted twice — **224 of 224 match by prompt, 0 of 224 by id**. So gating /practice would have deleted a feature the contract intends. The defect was the *disclosure*: answers are now withheld for graded rows, and the withholding **says so** rather than silently omitting.

Fail-closed on the R6 finding: nil catalog, load error **or stale snapshot** withholds every key, because a graded prompt the snapshot never saw would look “graded nowhere” to a literal index and be published.

R9 — What makes a QA document usable by someone who has never seen this?

Decision: Every scenario states its expected result **before** the tester acts; every known limitation is disclosed **in advance**; the build identifier is recorded.

Rationale. A step that says only “click X” makes the tester the oracle, and they have no basis to be one. And **a tester must never discover a known limitation as if it were a defect** — re-finding what we already know is the most expensive way to spend a QA team's time.

The documents must be regenerated, not patched (FR-020a).

Written at 12:30 against the pre-rebuild product, they describe 14 areas with tests and 0 lessons with content; live is 27 and 318 of 318. **A stale expected-result column manufactures false defects** and spends a tester's day disproving our own fixes.

R10 — What does the client see, and what is disclosed?

Decision: Disclose three facts up front (FR-020b) — 39 areas served against 42 authored, 12 carrying no test with the reason, and every score indeterminate.

Rationale. Each has a good reason, and **the reasons demonstrate the anti-bluff discipline rather than apologising for it.** A gap discovered live is worse than one disclosed first. The alternative — demo only the 27 areas with tests — is legitimate for a demo but leaves an undisclosed gap one click away in a catalogue showing 39.